

Assignment - 2

Sales Data Analysis Using SQL

Name: Suyash Saxena
Dataset: Sample Superstore
Database: PostgreSQL

Query Results

View first 10 records

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City
1	1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gite	Consumer	United States	Hendersonville
2	2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gite	Consumer	United States	Hendersonville
3	3	CA-2016-138688	6/12/2016	6/16/2016	Second Class	DV-13845	Darrin Van Huff	Corporate	United States	Los Angeles
4	4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale
5	5	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale
6	6	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles
7	7	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles
8	8	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles
9	9	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles
10	10	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles

Total number of records

total_rows	
1	9994

View table structure

	column_name	data_type
1	Row ID	bigint
2	Postal Code	bigint
3	Sales	double precision
4	Quantity	bigint
5	Discount	double precision
6	Profit	double precision
7	Customer Name	text
8	Segment	text
9	Country	text
10	City	text
11	State	text
12	Product Name	text
13	Region	text
14	Product ID	text
15	Category	text
16	Sub-Category	text
17	Order ID	text
18	Order Date	text
19	Ship Date	text
20	Ship Mode	text
21	Customer ID	text

Orders from West Region

"Row ID"	"Order ID"	"Order Date"	"Ship Date"	"Ship Mode"	"Customer ID"	"Customer Name"	"Segment"	"City"
3	CA-2016-138688	6/12/2016	6/16/2016	Second Class	DV-15045	Darrin Van Huff	Corporate	Unit
6	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	Unit
7	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	Unit
8	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	Unit
9	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	Unit
10	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	Unit
11	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	Unit
12	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	Unit
14	CA-2016-161389	12/5/2016	12/10/2016	Standard Class	IM-15070	Irene Maddox	Consumer	Unit
18	CA-2014-167164	5/13/2014	5/15/2014	Second Class	AG-10270	Alejandro Grove	Consumer	Unit
19	CA-2014-143336	8/27/2014	9/1/2014	Second Class	ZO-21925	Zuschuss Donatelli	Consumer	Unit
20	CA-2014-143336	8/27/2014	9/1/2014	Second Class	ZO-21925	Zuschuss Donatelli	Consumer	Unit
21	CA-2014-143336	8/27/2014	9/1/2014	Second Class	ZO-21925	Zuschuss Donatelli	Consumer	Unit
25	CA-2015-106320	9/25/2015	9/30/2015	Standard Class	EB-13870	Emily Burns	Consumer	Unit
26	CA-2016-121755	1/16/2016	1/20/2016	Second Class	EH-13945	Eric Hoffmann	Consumer	Unit
27	CA-2016-121755	1/16/2016	1/20/2016	Second Class	EH-13945	Eric Hoffmann	Consumer	Unit
43	CA-2016-101343	7/17/2016	7/22/2016	Standard Class	RA-19885	Ruben Ausman	Corporate	Unit
63	CA-2015-135545	11/24/2015	11/30/2015	Standard Class	KM-16720	Kunst Miller	Consumer	Unit
64	CA-2015-135545	11/24/2015	11/30/2015	Standard Class	KM-16720	Kunst Miller	Consumer	Unit
65	CA-2015-135545	11/24/2015	11/30/2015	Standard Class	KM-16720	Kunst Miller	Consumer	Unit
66	CA-2015-135545	11/24/2015	11/30/2015	Standard Class	KM-16720	Kunst Miller	Consumer	Unit
68	CA-2014-106376	12/5/2014	12/10/2014	Standard Class	BS-11590	Brendan Sweed	Corporate	Unit
69	CA-2014-106376	12/5/2014	12/10/2014	Standard Class	BS-11590	Brendan Sweed	Corporate	Unit

Technology Products

"Row ID"	"Order ID"	"Order Date"	"Ship Date"	"Ship Mode"	"Customer ID"	"Customer Name"	"Segment"	"City"
8	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	Unit
12	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	Unit
20	CA-2014-143336	8/27/2014	9/1/2014	Second Class	ZO-21925	Zuschuss Donatelli	Consumer	Unit
27	CA-2016-121755	1/16/2016	1/20/2016	Second Class	EH-13945	Eric Hoffmann	Consumer	Unit
36	CA-2016-117590	12/8/2016	12/10/2016	First Class	GH-14485	Gene Hale	Corporate	Unit
41	CA-2015-117415	12/27/2015	12/31/2015	Standard Class	SN-20710	Steve Nguyen	Home Office	Unit
42	CA-2017-120999	9/10/2017	9/15/2017	Standard Class	LC-16930	Linda Cazamias	Corporate	Unit
45	CA-2016-118255	3/11/2016	3/13/2016	First Class	ON-18715	Odella Nelson	Corporate	Unit
48	CA-2016-169194	6/20/2016	6/25/2016	Standard Class	LH-16900	Lena Hernandez	Consumer	Unit
49	CA-2016-169194	6/20/2016	6/25/2016	Standard Class	LH-16900	Lena Hernandez	Consumer	Unit
55	CA-2016-105816	12/11/2016	12/17/2016	Standard Class	JM-15265	Janet Molinari	Corporate	Unit
60	CA-2016-111682	6/17/2016	6/18/2016	First Class	TB-21055	Ted Butterfield	Consumer	Unit
63	CA-2015-135545	11/24/2015	11/30/2015	Standard Class	KM-16720	Kunst Miller	Consumer	Unit
69	CA-2014-106376	12/5/2014	12/10/2014	Standard Class	BS-11590	Brendan Sweed	Corporate	Unit
87	CA-2017-155558	10/26/2017	11/2/2017	Standard Class	PG-18895	Paul Gonzalez	Consumer	Unit
91	CA-2016-109886	9/17/2016	9/22/2016	Standard Class	JS-15685	Jim Sink	Corporate	Unit
101	CA-2016-158568	8/29/2016	9/2/2016	Standard Class	RB-19465	Rick Bensley	Home Office	Unit
104	US-2015-156867	11/13/2015	11/17/2015	Standard Class	LC-16870	Lena Cacioppo	Consumer	Unit
107	CA-2017-119904	11/23/2017	11/28/2017	Standard Class	JM-15250	Janet Martin	Consumer	Unit
108	CA-2017-119904	11/23/2017	11/28/2017	Standard Class	JM-15250	Janet Martin	Consumer	Unit
110	CA-2015-129476	10/15/2015	10/20/2015	Standard Class	PA-19860	Pete Armstrong	Home Office	Unit
124	CA-2016-103730	6/12/2016	6/15/2016	First Class	SC-20725	Steven Cartwright	Consumer	Unit
131	US-2017-164147	2/2/2017	2/5/2017	First Class	K-13585	Dorothy Wardle	Corporate	Unit

Orders with Sales above 1000

"Row ID"	"Order ID"	"Order Date"	"Ship Date"	"Ship Mode"	"Customer ID"	"Customer Name"	"Segment"	"City"
11	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	Unit
25	CA-2015-106320	9/25/2015	9/30/2015	Standard Class	EB-13870	Emily Burns	Consumer	Unit
28	US-2015-150630	9/17/2015	9/21/2015	Standard Class	TB-21520	Tracy Blumstein	Consumer	Unit
36	CA-2016-117590	12/8/2016	12/10/2016	First Class	GH-14485	Gene Hale	Corporate	Unit
55	CA-2016-105816	12/11/2016	12/17/2016	Standard Class	JM-15265	Janet Molinari	Corporate	Unit
68	CA-2014-106376	12/5/2014	12/10/2014	Standard Class	BS-11590	Brendan Sweed	Corporate	Unit
150	CA-2016-114489	12/5/2016	12/9/2016	Standard Class	JE-16165	Justin Ellison	Corporate	Unit
166	CA-2014-139892	9/8/2014	9/12/2014	Standard Class	BH-11140	Becky Martin	Consumer	Unit
168	CA-2014-139892	9/8/2014	9/12/2014	Standard Class	BH-11140	Becky Martin	Consumer	Unit
216	CA-2015-146262	1/2/2015	1/9/2015	Standard Class	VW-21775	Victoria Wilson	Corporate	Unit
245	CA-2014-131926	6/1/2014	6/6/2014	Second Class	DW-13480	Dianna Wilson	Home Office	Unit
248	CA-2014-131926	6/1/2014	6/6/2014	Second Class	DW-13480	Dianna Wilson	Home Office	Unit
252	CA-2016-145625	9/11/2016	9/17/2016	Standard Class	KC-16540	Kelly Collister	Consumer	Unit
263	US-2014-106992	9/19/2014	9/21/2014	Second Class	SB-20290	Sean Braxton	Corporate	Unit
264	US-2014-106992	9/19/2014	9/21/2014	Second Class	SB-20290	Sean Braxton	Corporate	Unit
282	US-2015-161991	9/26/2015	9/28/2015	Second Class	SC-20725	Steven Cartwright	Consumer	Unit
283	CA-2015-130898	11/2/2015	11/6/2015	Standard Class	JO-15280	Jas O'Carroll	Consumer	Unit
300	CA-2016-142545	10/28/2016	11/3/2016	Standard Class	JD-15895	Jonathan Doherty	Corporate	Unit
319	CA-2016-164973	11/4/2014	11/9/2014	Standard Class	NH-18445	Nathan Mautz	Home Office	Unit
354	CA-2016-129714	9/1/2016	9/3/2016	First Class	AB-10060	Adam Bellavance	Home Office	Unit
370	CA-2016-155516	10/21/2016	10/21/2016	Same Day	MK-17905	Michael Kennedy	Corporate	Unit
374	US-2014-119137	7/23/2014	7/27/2014	Standard Class	AG-10900	Arthur Gainer	Consumer	Unit
378	US-2017-134481	8/27/2017	9/1/2017	Standard Class	AR-10405	Allen Rosenblatt	Corporate	Unit

Orders with Profit above 100

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	City
1	2 CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States
2	14 CA-2016-161389	12/5/2016	12/10/2016	Standard Class	IM-15070	Irene Maddox	Consumer	United States
3	25 CA-2015-106320	9/25/2015	9/30/2015	Standard Class	EB-13870	Emily Burns	Consumer	United States
4	36 CA-2016-117590	12/8/2016	12/10/2016	First Class	GH-14485	Gene Hale	Corporate	United States
5	55 CA-2016-105816	12/11/2016	12/17/2016	Standard Class	JM-15265	Janet Molinar1	Corporate	United States
6	68 CA-2016-106376	12/5/2016	12/10/2016	Standard Class	BS-11590	Brendan Sweed	Corporate	United States
7	118 CA-2015-110457	3/2/2015	3/6/2015	Standard Class	DK-13090	Dave Kipp	Consumer	United States
8	145 CA-2017-155376	12/22/2017	12/27/2017	Standard Class	SG-20080	Sandra Glassco	Consumer	United States
9	148 CA-2016-114489	12/5/2016	12/9/2016	Standard Class	JE-16165	Justin Ellison	Corporate	United States
10	150 CA-2016-114489	12/5/2016	12/9/2016	Standard Class	JE-16165	Justin Ellison	Corporate	United States
11	160 CA-2016-114104	11/20/2016	11/24/2016	Standard Class	NP-18670	Nora Paige	Consumer	United States
12	183 CA-2014-158274	11/19/2014	11/24/2014	Second Class	RR-19675	Robert Marley	Home Office	United States
13	190 CA-2015-102281	10/12/2015	10/14/2015	First Class	MP-17470	Mark Packer	Home Office	United States
14	227 CA-2015-163955	8/9/2015	8/16/2015	Standard Class	DS-13180	David Smith	Corporate	United States
15	245 CA-2014-131926	6/1/2014	6/6/2014	Second Class	DW-13480	Dianna Wilson	Home Office	United States
16	248 CA-2014-131926	6/1/2014	6/6/2014	Second Class	DW-13480	Dianna Wilson	Home Office	United States
17	252 CA-2016-145625	9/11/2016	9/17/2016	Standard Class	KC-16540	Kelly Collister	Consumer	United States
18	254 CA-2016-146941	12/10/2016	12/13/2016	First Class	DL-13315	Delfina Latchford	Consumer	United States
19	271 CA-2017-163979	12/28/2017	1/2/2018	Second Class	KH-16690	Kristen Hastings	Corporate	United States
20	282 US-2015-161991	9/26/2015	9/28/2015	Second Class	SC-20725	Steven Cartwright	Consumer	United States
21	319 CA-2016-164973	11/4/2016	11/9/2016	Standard Class	NM-18445	Nathan Mautz	Home Office	United States
22	321 CA-2016-164973	11/4/2016	11/9/2016	Standard Class	NM-18445	Nathan Mautz	Home Office	United States
23	336 CA-2015-137946	9/1/2015	9/4/2015	Standard Class	3-13615	Doug Bickford	Consumer	United States

Orders with Discount greater than 30%

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	City
1	4 US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States
2	15 US-2015-118983	11/22/2015	11/26/2015	Standard Class	HP-14815	Harold Pawlan	Home Office	United States
3	16 US-2015-118983	11/22/2015	11/26/2015	Standard Class	HP-14815	Harold Pawlan	Home Office	United States
4	28 US-2015-150630	9/17/2015	9/21/2015	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States
5	29 US-2015-150630	9/17/2015	9/21/2015	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States
6	33 US-2015-150630	9/17/2015	9/21/2015	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States
7	37 CA-2016-117590	12/8/2016	12/10/2016	First Class	GH-14485	Gene Hale	Corporate	United States
8	39 CA-2015-117415	12/27/2015	12/31/2015	Standard Class	SN-20710	Steve Nguyen	Home Office	United States
9	76 US-2017-118038	12/9/2017	12/11/2017	First Class	KB-16600	Ken Brennan	Corporate	United States
10	77 US-2017-118038	12/9/2017	12/11/2017	First Class	KB-16600	Ken Brennan	Corporate	United States
11	79 US-2014-147606	11/26/2014	12/1/2014	Second Class	JE-15745	Joel Eaton	Consumer	United States
12	96 US-2017-109484	11/6/2017	11/12/2017	Standard Class	RB-19705	Roger Barcio	Home Office	United States
13	102 CA-2016-158568	8/29/2016	9/2/2016	Standard Class	RB-19465	Rick Bensley	Home Office	United States
14	106 US-2015-156867	11/13/2015	11/17/2015	Standard Class	LC-16870	Lena Cacioppo	Consumer	United States
15	119 US-2015-136476	4/5/2015	4/10/2015	Standard Class	GG-14650	Greg Guthrie	Corporate	United States
16	126 US-2014-134614	9/20/2014	9/25/2014	Standard Class	PF-19165	Philip Fox	Consumer	United States
17	127 US-2017-107272	11/5/2017	11/12/2017	Standard Class	TS-21610	Troy Staebel	Consumer	United States
18	131 US-2017-164147	2/2/2017	2/5/2017	First Class	DW-13585	Dorothy Wardle	Corporate	United States
19	166 CA-2014-139892	9/8/2014	9/12/2014	Standard Class	BM-11140	Becky Martin	Consumer	United States
20	170 CA-2014-139892	9/8/2014	9/12/2014	Standard Class	BM-11140	Becky Martin	Consumer	United States
21	175 US-2014-100853	9/14/2014	9/19/2014	Standard Class	JB-15400	Jennifer Braxton	Corporate	United States
22	177 US-2017-152366	4/21/2017	4/25/2017	Standard Class	SJ-20500	Shirley Jackson	Consumer	United States
23	199 US-2017-124303	7/6/2017	7/13/2017	Standard Class	4-14365	Fred Hopkins	Corporate	United States

Total Sales by Category

	Category	total_sales
1	Furniture	741999.7952999998
2	Office Supplies	719047.0320000029
3	Technology	836154.0329999996

Total Quantity by Region

	"Region"	total_quantity
1	South	6209
2	West	12266
3	East	10618
4	Central	8780

Average Profit by Segment

	"Segment"	avg_profit
1	Consumer	25.83687328067804
2	Corporate	30.45666688741729
3	Home Office	33.818664329781306

Average Discount by Category

	"Category"	avg_discount
1	Furniture	0.17392267798208297
2	Office Supplies	0.15728509790906708
3	Technology	0.1323226854358401

Total Sales by Ship Mode

	"Ship Mode"	total_sales
1	Standard Class	1358215.7429999826
2	Second Class	459193.5693999998
3	Same Day	128363.125
4	First Class	351428.4229000002

Products by Quantity Sold

	"Product Name"	total_quantity
1	Staples	215
2	Staple envelope	170
3	Easy-staple paper	150
4	Staples in misc. colors	86
5	KI Adjustable-Height Table	74
6	Avery Non-Stick Binders	71
7	Storex Dura Pro Binders	71
8	GBC Premium Transparent Covers with Diagonal Lined Pattern	67
9	Situations Contoured Folding Chairs, 4/Set	64
10	Staple-based wall hangings	62

Top 10 States by Sales

	"State"	total_sales
1	California	457687.631500001
2	New York	310876.2709999998
3	Texas	170188.04580000002
4	Washington	138641.26999999993
5	Pennsylvania	116511.91400000003
6	Florida	89473.708
7	Illinois	80166.10099999986
8	Ohio	78258.13599999993
9	Michigan	76269.61400000002
10	Virginia	70636.71999999999

Top 5 Customers by Quantity Purchased

	"Customer Name"	total_quantity
1	Jonathan Doherty	150
2	William Brown	146
3	John Lee	143
4	Paul Prost	138
5	Steven Cartwright	133

Top 10 Sub-Categories by Profit

	"Sub-Category"	total_profit
1	Copiers	55617.82490000001
2	Phones	44515.7306
3	Accessories	41936.63569999993
4	Paper	34053.56929999997
5	Binders	30221.763299999995
6	Chairs	26590.166300000026
7	Storage	21278.826399999998
8	Appliances	18138.005399999995
9	Furnishings	13059.143599999983
10	Envelopes	6964.176700000003

Sales by Year

	year	total_sales
1	2014	484247.4981000009
2	2015	470532.50899999985
3	2016	609205.5980000008
4	2017	733215.2551999999

Most Sold Products

	"Product Name"	total_quantity
1	Staples	215
2	Staple envelope	170
3	Easy-staple paper	150
4	Staples in misc. colors	86
5	KI Adjustable-Height Table	74
6	Avery Non-Stick Binders	71
7	Storex Dura Pro Binders	71
8	GBC Premium Transparent Covers with Diagonal Lined Pattern	67
9	Situations Contoured Folding Chairs, 4/Set	64
10	Staple-based wall hangings	62

Products with Highest Discounts

	"Product Name"	max_discount
1	Avery Binding System Hidden Tab Executive Style Index Sets	0.8
2	Recycled Easel Ring Binders	0.8
3	GBC Recycled Regency Composition Covers	0.8
4	XtraLife ClearVue Slant-D Ring Binders by Cardinal	0.8
5	Wilson Jones Legal Size Ring Binders	0.8
6	APC 7 Outlet Network SurgeArrest Surge Protector	0.8
7	Kensington 6 Outlet SmartSocket Surge Protector	0.8
8	Square Ring Data Binders, Rigid 75 Pt. Covers, 11" x 14-7/8"	0.8
9	Tuff Stuff Recycled Round Ring Binders	0.8
10	Acco 6 Outlet Guardian Premium Plus Surge Suppressor	0.8

Segment-wise Sales

	"Segment"	total_sales
1	Consumer	1161401.3449999888
2	Corporate	706146.3668000001
3	Home Office	429653.1485000003

Region-wise Quantity Sold

	"Region"	total_quantity
1	South	6209
2	West	12266
3	East	10618
4	Central	8780

Check Duplicate Order IDs

	Order ID	count
1	US-2017-160836	3
2	CA-2014-107811	2
3	CA-2015-131856	3
4	US-2017-118941	3
5	CA-2017-102197	3
6	CA-2016-153598	2
7	US-2017-128118	2
8	CA-2014-156244	6
9	CA-2017-162481	2
10	US-2017-169320	2
11	CA-2016-142405	3
12	CA-2016-139234	2
13	CA-2016-169957	3
14	CA-2015-154746	2
15	CA-2017-115994	2
16	CA-2015-156146	2
17	CA-2015-146255	4
18	CA-2017-100524	3
19	CA-2017-152499	2
20	CA-2017-132199	5
21	CA-2017-107517	2
22	CA-2014-168823	2
23	CA-2015-111325	2

Check Null Values

	null_records
1	0

Count Distinct Customers and Products

	total_customers	total_products
1	793	1862

Brief Insights

This project involved analyzing the Superstore sales dataset using SQL and PostgreSQL. The dataset was first loaded into a database and then explored using different SQL queries.

During the analysis, I used filtering techniques to examine specific records based on region, category, sales, profit, and discount values. Aggregate functions such as SUM(), AVG(), and COUNT() helped in understanding sales performance, customer activity, and product trends.

The analysis showed that sales performance varies across different categories, regions, and customer segments. By using sorting and ranking queries, it was possible to identify top-performing products, customers, and locations. Business-oriented queries also helped in understanding sales trends and customer purchasing patterns.

Data validation queries were performed to check record counts, identify duplicate entries, and verify the presence of missing values. These checks helped ensure the reliability of the dataset before drawing conclusions from it.

Overall, this assignment provided hands-on experience with SQL concepts such as SELECT, WHERE, GROUP BY, ORDER BY, LIMIT, aggregate functions, and data validation techniques. It also demonstrated how SQL can be used to extract meaningful insights from business data and support data-driven decision-making.